

MILESTONE CONTRACT

DOC-009 V1.0 (EXECUTIVE BI)

REAL-TIME OLAP ANALYTICS



Module 9: Executive Management Dashboard, 8-Module BI & Mobile Cockpit

Comprehensive 4-week milestone agreement, dedicated BI engineering team breakdown, cross-module OLAP data warehouse schema, real-time KPI engines, anomaly alert triggers, and mobile executive cockpit based on verified Business Analysis Report **DOC-009 v1.0** for **Popular Auto Spare & A/C Parts Co. W.L.L.**

TOTAL MODULE BUDGET

BD

1,523.636

PROJECT TIMELINE

4 Wks (1.0 Mo)

WEEKLY RESOURCE RATE

BD 380.909

PAYMENT TRANCHES

4 Gates (25% Ea)

Section Module 9: Functional Scope & Architecture

2.0 Breakdown (BA Ref: DOC-009)

FULL SPECIFICATION MAPPING

Derived directly from **DOC-009: Executive Management KPIs & BI Analytics Business Analysis Report**, Module 9 extracts operational feeds from all 8 upstream modules across 6 core management intelligence domains:

2.1 Commercial & Branch POS Performance Analytics

- **Real-Time Revenue:** Hourly branch sales tracking, daily revenue realization %, and monthly budget vs actual variance.
- **Profitability & Margins:** Gross margin realization % by PCode brand, category, branch, and salesperson commission tier.
- **Average Transaction Value (ATV):** Retail basket size analytics, items per invoice ratio, and discount variance index.
- **Showroom Footfall & Cash Drawer:** Hourly showroom traffic heatmaps and cash drawer variance tracking.

2.2 Dynamic Inventory Health & Supply Chain Intelligence

- **Dead Stock Risk Index:** Automated classification of non-moving PCodes (>90, >180, >365 days) with tied-up capital calculations.
- **Turnover Ratio & Fast Movers:** Stock turnover velocity by manufacturer brand (Toyota, Nissan, Denso, Sanden) and reorder forecasting.
- **Inter-Branch Transit Logistics:** Tracking stock transfer lead times and in-transit inventory value across warehouses.
- **Inventory Accuracy Index:** Cycle count variance tracking, physical vs book stock discrepancies, and shrinkage monitoring.

2.3 Financial Liquidity & Executive Cash Flow Cockpit

- **Daily Working Capital:** Consolidated cash and bank balances across all corporate accounts with 30-day runway projection.

- **Accounts Receivable Aging:** Risk-weighted AR debtor aging buckets (Current, 30, 60, 90+ days) and DSO metrics.
- **Supplier AP Liabilities:** Supplier payment due calendar, early settlement discount opportunities, and credit limit utilization.
- **VAT Compliance Forecast:** Real-time Kingdom of Bahrain 10% VAT output vs input tax liability forecasting.

2.4 Multi-Branch Operations & Workforce Productivity

- **Fleet Logistics & TCO:** Delivery vehicle maintenance cost per kilometer, fuel consumption index, and dispatch efficiency.
- **Workforce Productivity:** Revenue per employee ratio, overtime cost vs sales realization, and biometric punctuality rate.
- **Branch Operating Expenses (OPEX):** Utility costs, rental amortization, and branch administrative spend vs revenue benchmarks.
- **Procurement Turnaround:** Vendor Purchase Order fulfillment lead time and supplier price variance index.

2.5 Cross-Module OLAP Aggregation & Anomaly Triggers

- **High-Speed OLAP Layer:** Dedicated data aggregation schema extracting hourly transactional feeds without impacting live OLTP speed.
- **Automated Anomaly Detection:** Machine-assisted triggers for sudden margin drops, abnormal discounts, and uncharacteristic stock adjustments.
- **Executive Notification Engine:** Automated daily digest summaries and high-priority threshold alerts via Email and SMS.
- **Custom KPI Builder:** Flexible metric formula designer for executive management to create tailored ad-hoc indices.

2.6 Executive Mobile Cockpit & Role-Based Governance

- **Responsive Executive Cockpit:** Tailored dark/light mode mobile interface for C-suite leadership (MD, GM, Financial Controller).
- **Drill-Down Analytics:** Interactive drill-down capabilities from high-level consolidated cards down to branch and transaction level.
- **Role-Based Access Governance:** Strict encryption and granular permission layers ensuring executive confidentiality.

- **Export & Reporting:** Instant one-click export of executive BI summaries to PDF board presentations and Excel workbooks.

Section 3.0

Dedicated BI Engineering Resource Allocation
(4 Weeks / 1.0 Month)

COST & ALLOCATION BREAKDOWN

To build the multi-source data aggregation pipelines and responsive executive dashboard UI, SaNDS Lab allocates a specialized 3-member BI engineering team:

ROLE & EXPERTISE	FTE ALLOCATION	WEEKLY RATE	4-WEEK TOTAL (BD)	PRIMARY DELIVERABLES
Lead Data Warehouse & BI Architect	0.50 FTE	BD 152.364	BD 609.456	OLAP data schema, cross-module ETL pipeline architecture, anomaly alert algorithms
Senior Data Visualization & UI Engineer	1.00 FTE	BD 152.364	BD 609.456	Interactive dashboard UI, responsive mobile cockpit, chart rendering engines, drill-down logic
Total Dedicated Team Allocation	2.00 FTE	BD 380.909 / Wk	BD 1,523.636	4 Working Weeks (20 Days)

ROLE & EXPERTISE	FTE ALLOCATION	WEEKLY RATE	4-WEEK TOTAL (BD)	PRIMARY DELIVERABLES
ETL Pipeline & DB Optimization Specialist	0.50 FTE	BD 76.181	BD 304.724	Database query indexing, high-concurrency caching, automated hourly aggregation jobs
Total Dedicated Team Allocation	2.00 FTE	BD 380.909 / Wk	BD 1,523.636	4 Working Weeks (20 Days)

Section 4.0

Detailed Milestone Breakdown & Deliverables (4-Week Roadmap)

SPRINT EXECUTION PLAN

The implementation is organized into 4 sequential weekly milestones (25.00% each), delivering continuous BI intelligence across all 8 operational modules:

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Milestone 9.1: Data Warehouse OLAP Aggregation Schema & Cross-Module ETL Pipelines

WEEK 1 (1 WK)

Architecting the high-speed Data Warehouse OLAP layer that extracts, normalizes, and aggregates raw transactional data from all 8 upstream ERP modules without degrading live database performance.

✓ OLAP Data Warehouse Schema Implemented

✓ 8-Module Automated ETL Data Extraction Pipes

- ✓ Sub-Second Aggregation Query Optimization
- ✓ Hourly Incremental Sync Pipeline Active

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Milestone 9.2: Real-Time Executive KPI Metric Calculation Engines & Anomaly Alert Triggers

WEEK 2 (1 WK)

Implementing the core analytical math engine powering commercial POS metrics, dead stock velocity, AR aging risk scores, gross margin realization, and threshold-based anomaly alerts.

- ✓ Commercial POS & Gross Margin Formula Engine
- ✓ Dead Stock Aging Index & Turnover Calculation
- ✓ AR/AP Working Capital & Cash Runway Engine
- ✓ Automated Email & SMS Anomaly Alert Service

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Milestone 9.3: Visual Dashboard UI, Interactive Charts & Drill-Down Analytical Cockpit

WEEK 3 (1 WK)

Designing and rendering the rich web-based executive dashboard interface with interactive charts, branch comparative gauges, PCode revenue heatmaps, and transaction drill-down consoles.

- ✓ C-Suite Interactive Visual Charts & Heatmaps
- ✓ Multi-Branch Real-Time Comparison Matrix
- ✓ Click-to-Drill Down Transaction Inspector
- ✓ One-Click PDF Board Summary Report Exporter

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Milestone 9.4: Mobile Executive Cockpit, Role-Based Access Governance & Final UAT

WEEK 4 (1 WK)

Optimization of mobile-responsive touch cockpit for smartphones and tablets, granular RBAC executive permission governance, security auditing, and final C-suite UAT sign-off.

- ✓ Mobile Touch-Optimized C-Suite Cockpit UI
- ✓ End-to-End Cross-Module Data Verification
- ✓ Granular Role-Based Data Visibility Rules
- ✓ Executive Production Sign-Off & Acceptance

Section 5.0 Milestone Payment Structure & Invoicing Schedule (4-Week Model)

VERIFIABLE MILESTONE BILLING

Invoicing is strictly tied to verifiable technical deliverables and stakeholder sign-off across 4 equal weekly milestone gates:

GATE	MILESTONE DELIVERABLE FOCUS	TIMELINE	% FEE	INVOICE AMOUNT (BD)	VERIFICATION CRITERIA
Gate 9.1	OLAP Schema, Data Warehouse & ETL Pipelines	End of Wk 1	25.00%	BD 380.909	Cross-module ETL extraction validated against live DB
Gate 9.2	Real-Time KPI Calculation Engines & Anomaly Alerts	End of Wk 2	25.00%	BD 380.909	POS, dead stock, and AR formulas verified by Finance
Total Module 9 Professional Investment			100.00%	BD 1,523.636	4 Equal Milestone Tranches

GATE	MILESTONE DELIVERABLE FOCUS	TIMELINE	% FEE	INVOICE AMOUNT (BD)	VERIFICATION CRITERIA
Gate 9.3	Visual Dashboard UI & Drill-Down Analytics	End of Wk 3	25.00%	BD 380.909	Interactive web UI charts & drill-downs verified
Gate 9.4	Mobile Executive Cockpit, RBAC & Final Sign-Off	End of Wk 4	25.00%	BD 380.909	Mobile cockpit approved & final executive acceptance
Total Module 9 Professional Investment			100.00%	BD 1,523.636	4 Equal Milestone Tranches

Section 6.0 Working Calendar, SLA, 15–Day Grace Period & Legal Terms

GOVERNANCE FRAMEWORK

Bahrain Working Calendar & Schedule

Standard working schedule is Sunday through Thursday (5 business days / 40 hours per week). Official Kingdom of Bahrain statutory public holidays are observed and accounted for in the 4–week timeline.

15–Day Quality Review & Grace Period

Following completion of each milestone gate, Popular Auto Parts is granted a 15–calendar-day review window to validate deliverables. If no written defects are reported within 15 days, milestone acceptance is confirmed.

Section 7.0 Stakeholder Authorization & Digital Sign-Off Console

EXECUTIVE ACCEPTANCE

By electronically signing below, the authorized executive stakeholders formally accept the scope, deliverables, pricing, and milestone schedule for **Module 9: Executive Management Dashboard, 8–Module BI & Mobile Cockpit**.

Implementation Partner

✓ SIGNED

⌚ PENDING SIGNATURE

SaNDS Lab / UniGlobal Consulting

 SaNDS Signature  Signature Pending

Signer:

Title:

Signed Date:

 Sign for SaNDS Lab

Client Organization

✓ SIGNED

⌚ PENDING SIGNATURE

Popular Auto Spare & A/C Parts Co. W.L.L.

 Client Signature  Signature Pending

Signer:

Title:

Signed Date:

 Sign for Popular Auto Parts

 Finalize & Lock Milestone Acceptance

✓ Document Accepted and Finalized into Master ERP Registry.